Reconstruction Metric - GPU Optimized

2026-05-19

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0.1 1. Setup e Imports

```python
import numpy as np
import pandas as pd
import sys
import os
import subprocess
import time
import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt

from pathlib import Path
from tqdm import tqdm
from sae.tools.experiment_utils import get_experiment_dir, get_metrics_dir, get_report_name
from sae.tools.naming_utils import get_checkpoint_dir, get_model_name, get_extras_id

project_root = Path('../../..').resolve()
sys.path.insert(0, str(project_root))

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Device: {device}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
    print(f"Memoria: {torch.cuda.get_device_properties(0).total_memory / 1e9:.2f} GB")
```

```
Device: cuda
GPU: NVIDIA GeForce RTX 5060
Memoria: 8.55 GB
```

```python
# Configuración para visualización en PDF
pd.set_option('display.max_colwidth', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 100)
```

```python
np.set_printoptions(linewidth=100, edgeitems=3)

os.environ['COLUMNS'] = '100'

print(" Configuración para PDF lista")
```

```
 Configuración para PDF lista
```

```python
NAMING_META = {
    'variante': 'topk',
    'tecnica': 'batch-topk',
    'capa': 6,
    'juegos': 100000,
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',
    'experiment_base_path': os.path.abspath('../../../experiments'),
    'extras': {
        'expansion': 16,
        'k': 81,
        'epochs': 1000,
        'patience': 20,
        'batch_size_tokens': 1024,
    }
}

print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f'  {key}: {value}')
```

```

 Metadata de naming:
  variante: topk
  tecnica: batch-topk
  capa: 6
  juegos: 100000
  base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
  experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
  extras: {'expansion': 16, 'k': 81, 'epochs': 1000, 'patience': 20, 'batch_size_tokens': 1024}
```

```python
class BatchTopKSAE(nn.Module):
    def __init__(self, d_in: int, d_sae: int, k: int, k_aux: int = 512):
        super().__init__()
        self.d_in  = d_in
        self.d_sae = d_sae
        self.k     = k
        self.k_aux = k_aux

        self.W_enc = nn.Parameter(torch.empty(d_in, d_sae))
        self.W_dec = nn.Parameter(torch.empty(d_sae, d_in))
        self.b_dec = nn.Parameter(torch.zeros(d_in))

        nn.init.kaiming_uniform_(self.W_enc)
        nn.init.kaiming_uniform_(self.W_dec)
        self.W_dec.data = self.W_dec.data / self.W_dec.data.norm(dim=-1, keepdim=True)
        self.register_buffer('num_batches_not_active', torch.zeros(d_sae))

    def encode(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
        x_norm, _, _ = self.preprocess_input(x)
        x_cent   = x_norm - self.b_dec
        pre_acts = F.relu(x_cent @ self.W_enc)
        return self._topk_activation(pre_acts, threshold=threshold)
```

```python
    def _topk_activation(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
        if threshold is None:
            acts_topk = torch.topk(x.flatten(), self.k * x.shape[0], dim=-1)
            return (
                torch.zeros_like(x.flatten())
                .scatter(-1, acts_topk.indices, acts_topk.values)
                .reshape(x.shape)
            )
        return torch.where(x > threshold, x, torch.zeros_like(x))

    def decode(self, hidden: torch.Tensor) -> torch.Tensor:
        return hidden @ self.W_dec + self.b_dec

    def preprocess_input(self, x: torch.Tensor):
        x_mean = x.mean(dim=-1, keepdim=True)
        x = x - x_mean
        x_std = x.std(dim=-1, keepdim=True)
        x = x / (x_std + 1e-5)
        return x, x_mean, x_std

    def postprocess_output(self, x_reconstruct, x_mean, x_std):
        return x_reconstruct * x_std + x_mean

    def forward(self, x: torch.Tensor, threshold=None):
        x_norm, x_mean, x_std = self.preprocess_input(x)
        x_cent   = x_norm - self.b_dec
        pre_acts = F.relu(x_cent @ self.W_enc)
        hidden   = self._topk_activation(pre_acts, threshold=threshold)
        recon    = self.postprocess_output(self.decode(hidden), x_mean, x_std)
        return recon, hidden, pre_acts

print("BatchTopKSAE definida")
```

```
BatchTopKSAE definida
```

0.2 2. Cargar Datos

```python
# Cargar activaciones
activations_path = project_root / "activations" / "data" / "layer6_2000games.npy"
activations = np.load(activations_path)

print(f"Activaciones del modelo:")
print(f"  Shape: {activations.shape}")
print(f"  Memoria: {activations.nbytes / (1024**2):.2f} MB")
```

```
Activaciones del modelo:
  Shape: (118000, 512)
  Memoria: 230.47 MB
```

```python
# Cargar ground truth de BSPs
bsp_gt_path = project_root / "metrics" / "02_data" / "bsp_ground_truth_2000games.npy"
bsp_names_path = project_root / "metrics" / "02_data" / "bsp_ground_truth_2000games.names.npy"

bsp_ground_truth = np.load(bsp_gt_path)
bsp_names = np.load(bsp_names_path, allow_pickle=True)

print(f"\nGround truth BSPs:")
print(f"  Shape: {bsp_ground_truth.shape}")
print(f"  Total BSPs: {len(bsp_names)}")
```

```
Ground truth BSPs:
  Shape: (118000, 326)
  Total BSPs: 326
```

0.3 3. Cargar SAE y Extraer Features

```python
# Configuración del SAE
input_dim = 512
hidden_dim = 8192

# Construir ruta del modelo usando naming_utils
checkpoint_dir = get_checkpoint_dir(
    NAMING_META['base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos']
)
extras_id = get_extras_id(checkpoint_dir, NAMING_META['extras']) if NAMING_META.get('extras') else None

model_filename = get_model_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'best',
    extras_id=extras_id
)
model_path = checkpoint_dir / model_filename

# Cargar modelo BatchTopK
sae = BatchTopKSAE(d_in=input_dim, d_sae=hidden_dim, k=NAMING_META['extras']['k']).to(device)

checkpoint = torch.load(model_path, map_location=device, weights_only=False)
sae.load_state_dict(checkpoint['model_state_dict'])
sae.eval()

print(f"SAE cargado:")
print(f"  Path: {model_path}")
print(f"  Input: {input_dim}, Hidden: {hidden_dim}, k: {NAMING_META['extras']['k']}")
print(f"  Expansion: {hidden_dim/input_dim}x")
```

```
SAE cargado:
  Path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk\sae_batch-topk_topk\100000games\sae_to
topk_l6_100000g_ex2_best.pt
  Input: 512, Hidden: 8192, k: 81
  Expansion: 16.0x
```

```python
# Extraer features del SAE en GPU
print("Extrayendo features del SAE...")
activations_tensor = torch.from_numpy(activations).float().to(device)

# Estimar threshold con mini-batch
with torch.no_grad():
    sample = activations_tensor[:4096]
    x_s, _, _ = sae.preprocess_input(sample)
    pre_s = F.relu((x_s - sae.b_dec) @ sae.W_enc)
    threshold = torch.topk(pre_s.flatten(), sae.k * 4096).values.min().item()
print(f"Threshold estimado: {threshold:.6f}")
```

```
# Encoding completo con operación in-place
with torch.no_grad():
    x_norm, _, _ = sae.preprocess_input(activations_tensor)
    x_cent = x_norm - sae.b_dec
    pre_acts = F.relu(x_cent @ sae.W_enc)
    pre_acts.mul_(pre_acts > threshold)
    sae_features = pre_acts

del activations_tensor, x_norm, x_cent
torch.cuda.empty_cache()

print(f"\nFeatures SAE:")
print(f"  Shape: {sae_features.shape}")
print(f"  Device: {sae_features.device}")
print(f"  Sparsity: {(sae_features == 0).float().mean():.2%}")
print(f"  Activaciones promedio: {(sae_features > 0).sum(dim=1).float().mean():.1f}")
```

```
Extrayendo features del SAE...
Threshold estimado: 0.911244

Features SAE:
  Shape: torch.Size([118000, 8192])
  Device: cuda:0
  Sparsity: 99.01%
  Activaciones promedio: 80.9
```

0.4 4. Split Train/Test y Filtrar BSPs de Piezas

```
# Split 50/50:
n_moves = 59
n_games = 2000  # número de partidas para prueba de métricas
train_games = n_games // 2
split_idx = train_games * n_moves

sae_features_train = sae_features[:split_idx]
sae_features_test  = sae_features[split_idx:]

print(f"Total partidas: {n_games}")
print(f"Train: {train_games} partidas ({split_idx} activaciones)")
print(f"Test:  {n_games - train_games} partidas ({len(activations) - split_idx} activaciones)")
print(f"Train set: {sae_features_train.shape}")
print(f"Test set:  {sae_features_test.shape}")
```

```
Total partidas: 2000
Train: 1000 partidas (59000 activaciones)
Test:  1000 partidas (59000 activaciones)
Train set: torch.Size([59000, 8192])
Test set:  torch.Size([59000, 8192])
```

```
# ============================================
# CONFIGURACIÓN DE FILTRADO DE BSPs
# ============================================
USE_PLAYER    = False  # BSPs perspectiva del jugador (1 y 2)
USE_ABSOLUTE = True    # BSPs absolutas negro/blanco (B y W)
USE_STRATEGIC = False  # BSPs especiales (BSP_)

bsp_pieces_indices = []
bsp_pieces_names = []
```

```python
for i, name in enumerate(bsp_names):
    include = False
    if USE_PLAYER and len(name) == 6 and name.startswith('BSP') and (name.endswith('1') or name.endswith('2')):
        include = True
    if USE_ABSOLUTE and (name.endswith('B') or name.endswith('W')):
        include = True
    if USE_STRATEGIC and name.startswith('BSP_'):
        include = True
    if include:
        bsp_pieces_indices.append(i)
        bsp_pieces_names.append(name)

bsp_pieces_indices = np.array(bsp_pieces_indices)

# Para reconstruction necesitas split train/test
bsp_gt_train_pieces = torch.from_numpy(bsp_ground_truth[:split_idx, bsp_pieces_indices]).bool().to(device)
bsp_gt_test_pieces  = torch.from_numpy(bsp_ground_truth[split_idx:, bsp_pieces_indices]).bool().to(device)

# Convertir a tensor en GPU
# Identificador del filtrado (para naming de archivos de salida)
parts = []
if USE_PLAYER: parts.append("player")
if USE_ABSOLUTE: parts.append("absolute")
if USE_STRATEGIC: parts.append("strategic")
filter_suffix = "_".join(parts) if parts else "none"

print(f"BSPs seleccionadas para Reconstruction:")
print(f"  Player: {USE_PLAYER} | Absolute: {USE_ABSOLUTE} | Strategic: {USE_STRATEGIC}")
print(f"  Total: {len(bsp_pieces_indices)}")
print(f"  Train shape: {bsp_gt_train_pieces.shape}")
print(f"  Test shape:  {bsp_gt_test_pieces.shape}")
print(f"  Device: {bsp_gt_train_pieces.device}")
print(f"  Primeras 5: {', '.join(bsp_pieces_names[:5])}")
print(f"  Últimas 5: {', '.join(bsp_pieces_names[-5:])}")
print(f"  Filter suffix: {filter_suffix}")
```

```
BSPs seleccionadas para Reconstruction:
  Player: False | Absolute: True | Strategic: False
  Total: 128
  Train shape: torch.Size([59000, 128])
  Test shape:  torch.Size([59000, 128])
  Device: cuda:0
  Primeras 5: BSPA1B, BSPA1W, BSPA2B, BSPA2W, BSPA3B
  Últimas 5: BSPH6W, BSPH7B, BSPH7W, BSPH8B, BSPH8W
  Filter suffix: absolute
```

0.5 5. Funciones GPU-Optimizadas

```python
def fast_precision_score_gpu(y_true, y_pred, eps=1e-8):
    """
    Precisión vectorizada en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions, n_candidates) booleano

    Returns:
        precision_scores: Tensor (n_candidates,)
    """
    y_true_expanded = y_true.unsqueeze(1)
```

```python
    tp = (y_true_expanded & y_pred).sum(dim=0).float()
    fp = (~y_true_expanded & y_pred).sum(dim=0).float()

    precision = tp / (tp + fp + eps)

    return precision


def fast_f1_score_gpu(y_true, y_pred, eps=1e-8):
    """
    F1 score vectorizado en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions,) booleano

    Returns:
        f1: Float
    """
    tp = (y_true & y_pred).sum().float()
    fp = (~y_true & y_pred).sum().float()
    fn = (y_true & ~y_pred).sum().float()

    precision = tp / (tp + fp + eps)
    recall = tp / (tp + fn + eps)

    f1 = 2 * (precision * recall) / (precision + recall + eps)

    return f1.item()

print(" Funciones GPU definidas")
```

```
 Funciones GPU definidas
```

0.6 6. Fase 1: Identificar Features de Alta Precisión (GPU)

Para cada BSP, encontrar features con precisión 0.95 en el train set.

```python
def identify_high_precision_features_gpu(
    sae_features_train,
    bsp_gt_train,
    f_max,
    precision_threshold=0.95,
    significance_threshold=1,
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    feature_batch_size=512,
    verbose=True
):
    """
    Para cada (threshold t, feature f, BSP b): determina si f es clasificador
    de alta precisión para b al threshold t.

    Criterios (igual que implementación oficial):
        - precision(f, b, t) >= precision_threshold   (default 0.95)
        - on_count(f, t)     >= significance_threshold (default 10)

    Args:
        sae_features_train:    Tensor (P, n_features) GPU
        bsp_gt_train:          Tensor (P, n_bsps) GPU, bool
        f_max:                 Tensor (n_features,) GPU - calculado FUERA para
```

```
                              garantizar identidad exacta con Fase 2
        precision_threshold:   float, default 0.95
        significance_threshold: int, mín. activaciones requeridas (default 10)
        thresholds:            lista de fracciones t   [0, 1)
        feature_batch_size:    int, features por mini-batch interno

    Returns:
        hp_mask_TFB:            BoolTensor (T, n_active, n_bsps)
        active_feature_indices: LongTensor (n_active,)
    """
    n_positions, n_features = sae_features_train.shape
    n_bsps = bsp_gt_train.shape[1]
    device = sae_features_train.device

    thresholds_T = torch.tensor(thresholds, device=device)
    n_thresholds = len(thresholds_T)

    # Features vivas (f_max > 0)
    active_feature_indices = (f_max > 0).nonzero(as_tuple=True)[0]
    n_active = len(active_feature_indices)

    if verbose:
        print("=" * 60)
        print("FASE 1: IDENTIFICAR FEATURES DE ALTA PRECISIÓN")
        print("=" * 60)
        print(f"Precision threshold:     {precision_threshold}")
        print(f"Significance threshold:  >= {significance_threshold} activaciones")
        print(f"Features activas:        {n_active}/{n_features} ({n_active/n_features*100:.1f}%)")
        print(f"BSPs:                    {n_bsps}")
        print(f"Thresholds:              {thresholds}")
        print("=" * 60)

    # Thresholds absolutos: thresh_TF[t, f] = t * f_max[f]
    active_f_max  = f_max[active_feature_indices]          # (n_active,)
    thresholds_TF = thresholds_T[:, None] * active_f_max   # (T, n_active)

    # Acumuladores sobre todo el train set
    on_count_TF  = torch.zeros(n_thresholds, n_active, device=device)
    tp_count_TFB = torch.zeros(n_thresholds, n_active, n_bsps, device=device)

    bsp_float_PB = bsp_gt_train.float()  # (P, B)

    n_iters = (n_active + feature_batch_size - 1) // feature_batch_size

    for fi in tqdm(range(n_iters), desc="Fase 1"):
        f_start = fi * feature_batch_size
        f_end   = min(f_start + feature_batch_size, n_active)

        global_idx = active_feature_indices[f_start:f_end]
        acts_PF    = sae_features_train[:, global_idx]     # (P, bs)
        thresh_TF  = thresholds_TF[:, f_start:f_end]      # (T, bs)

        # active_TFP[t, f, p] = True si acts[p,f] > thresh[t,f]
        active_TFP = acts_PF.T.unsqueeze(0) > thresh_TF.unsqueeze(2)  # (T, bs, P)

        # Conteo de activaciones por (threshold, feature)
        on_count_TF[:, f_start:f_end] += active_TFP.sum(dim=2).float()

        # True positives: f activa Y BSP verdadera
        # tp[t, f, b] = Σ_p active[t,f,p] * bsp[p,b]
        tp_count_TFB[:, f_start:f_end, :] += torch.einsum(
            'tfp,pb->tfb', active_TFP.float(), bsp_float_PB
```

```python
        )

    # Precisión: tp / on_count
    eps = 1e-8
    precision_TFB = tp_count_TFB / (on_count_TF.unsqueeze(2) + eps)  # (T, F, B)

    # Máscara: alta precisión AND significancia suficiente
    sig_mask_TF1 = (on_count_TF >= significance_threshold).unsqueeze(2)  # (T, F, 1)
    hp_mask_TFB  = (precision_TFB >= precision_threshold) & sig_mask_TF1  # (T, F, B)

    if verbose:
        hp_per_t = hp_mask_TFB.sum(dim=(1, 2))  # (T,)
        hp_per_b = hp_mask_TFB.sum(dim=1)        # (T, B)
        best_t   = hp_per_t.argmax().item()
        print(f"\nPares (feature, BSP) de alta precisión por threshold:")
        for i, t in enumerate(thresholds):
            marker = " ← más pares" if i == best_t else ""
            print(f"  t={t:.1f}: {hp_per_t[i].item():5d} pares{marker}")
        print(f"\nFeatures HP por BSP en t={thresholds[best_t]:.1f}:")
        print(f"  Media:             {hp_per_b[best_t].float().mean():.1f}")
        print(f"  Máx:               {hp_per_b[best_t].max().item()}")
        print(f"  BSPs sin features: {(hp_per_b[best_t] == 0).sum().item()}")

    return hp_mask_TFB, active_feature_indices

print("  Fase 1 rediseñada: umbral de significancia ( 10) + estructura (T, F, B)")
```

```
  Fase 1 rediseñada: umbral de significancia ( 10) + estructura (T, F, B)
```

```python
THRESHOLDS = torch.arange(0.0, 1.05, 0.05).tolist()


# f_max calculado UNA SOLA VEZ aquí - se pasa a Fase 1 y Fase 2
f_max = sae_features_train.max(dim=0)[0]  # (n_features,)
print(f"f_max calculado: {(f_max > 0).sum().item()} features activas de {f_max.shape[0]}")

start_time = time.time()

hp_mask_TFB, active_feature_indices = identify_high_precision_features_gpu(
    sae_features_train,
    bsp_gt_train_pieces,
    f_max,
    precision_threshold=0.95,
    significance_threshold=1,
    thresholds=THRESHOLDS,
    feature_batch_size=512,
    verbose=True
)

phase1_time = time.time() - start_time

print(f"\nTiempo Fase 1: {phase1_time:.2f} seg")
print(f"hp_mask_TFB shape: {hp_mask_TFB.shape}  (T, n_active, n_bsps)")
```

```
f_max calculado: 7132 features activas de 8192
============================================================
FASE 1: IDENTIFICAR FEATURES DE ALTA PRECISIÓN
============================================================
Precision threshold:     0.95
Significance threshold:  >= 1 activaciones
Features activas:        7132/8192 (87.1%)
```

```
BSPs:                    128
Thresholds:              [0.0, 0.05000000074505806, 0.10000000149011612, 0.15000000596046448, 0.20000000298023224, 0.25
============================================================

Fase 1: 100%|     | 14/14 [00:00<00:00, 14.16it/s]


Pares (feature, BSP) de alta precisión por threshold:
  t=0.0: 19514 pares
  t=0.1: 19514 pares
  t=0.1: 19756 pares
  t=0.2: 20495 pares
  t=0.2: 21895 pares
  t=0.2: 23818 pares
  t=0.3: 26095 pares
  t=0.3: 28495 pares
  t=0.4: 31106 pares
  t=0.5: 33840 pares
  t=0.5: 36681 pares
  t=0.6: 39404 pares
  t=0.6: 42574 pares
  t=0.6: 45610 pares
  t=0.7: 49380 pares
  t=0.8: 53449 pares
  t=0.8: 58783 pares
  t=0.9: 65813 pares
  t=0.9: 78294 pares
  t=0.9: 96968 pares ← más pares
  t=1.0:     0 pares

Features HP por BSP en t=0.9:
  Media:            757.6
  Máx:              4048
  BSPs sin features: 0

Tiempo Fase 1: 34.25 seg
hp_mask_TFB shape: torch.Size([21, 7132, 128])  (T, n_active, n_bsps)
```

0.7 7. Fase 2: Reconstruir Tableros en Test Set (GPU)

```python
def reconstruct_boards_gpu(
    sae_features_test,
    bsp_gt_test,
    hp_mask_TFB,
    active_feature_indices,
    f_max,
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    verbose=True
):
    """
    Fase 2: barrido simultáneo de thresholds T → toma el mejor F1.

    Para cada threshold t:
      1. Binarizar features de test: active[p, f] = acts[p, f] > t * f_max[f]
      2. Predicción por (posición, BSP):
           predict[p, b] = OR_f ( active[p,f] AND hp_mask[t, f, b] )
           implementado como: (active_PF @ hp_FB) > 0
      3. F1 global micro-averaged (acumula TP/FP/FN sobre todas las posiciones y BSPs)

    Retorna el max F1 sobre todos los thresholds (Eq. 5 del paper: max_t).
```

```
    Args:
        sae_features_test:      Tensor (P, n_features) GPU
        bsp_gt_test:            Tensor (P, n_bsps) GPU, bool
        hp_mask_TFB:            BoolTensor (T, n_active, n_bsps) de Fase 1
        active_feature_indices: LongTensor (n_active,) de Fase 1
        f_max:                  Tensor (n_features,) GPU - MISMO objeto que Fase 1
        thresholds:             lista de fracciones (debe coincidir con Fase 1)

    Returns:
        best_f1:          float, max F1 sobre thresholds
        best_threshold:   float, threshold óptimo
        f1_per_threshold: np.array (T,), F1 por threshold
    """
    device = sae_features_test.device
    n_thresholds = len(thresholds)
    thresholds_T = torch.tensor(thresholds, device=device)

    # Activaciones del test solo para features activas
    test_active_PF = sae_features_test[:, active_feature_indices]  # (P, n_active)
    active_f_max   = f_max[active_feature_indices]                  # (n_active,)

    gt_PB = bsp_gt_test.bool()  # (P, B)
    eps   = 1e-8

    f1_scores   = torch.zeros(n_thresholds, device=device)
    precision_T = torch.zeros(n_thresholds, device=device)
    recall_T    = torch.zeros(n_thresholds, device=device)

    if verbose:
        print("=" * 60)
        print("FASE 2: RECONSTRUCCIÓN - BARRIDO DE THRESHOLDS")
        print("=" * 60)
        print(f"{'t':>5}  {'F1':>8}  {'P':>8}  {'R':>8}  {'TP':>8}  {'FP':>8}  {'FN':>8}")
        print("-" * 60)

    for t_idx in range(n_thresholds):
        t = thresholds[t_idx]

        # Threshold absoluto por feature: t * f_max[f]
        abs_thresh_F = thresholds_T[t_idx] * active_f_max  # (n_active,)

        # Binarizar features en test
        active_PF = test_active_PF > abs_thresh_F.unsqueeze(0)  # (P, n_active)

        # predict[p, b] = any( active[p,f] AND hp[t,f,b] )
        # = (active_PF.float() @ hp_mask[t].float()) > 0
        hp_FB          = hp_mask_TFB[t_idx].float()              # (n_active, B)
        predictions_PB = (active_PF.float() @ hp_FB) > 0         # (P, B)

        # Métricas globales micro-averaged
        tp = (predictions_PB &  gt_PB).sum().float()
        fp = (predictions_PB & ~gt_PB).sum().float()
        fn = (~predictions_PB & gt_PB).sum().float()

        p  = tp / (tp + fp + eps)
        r  = tp / (tp + fn + eps)
        f1 = 2 * p * r / (p + r + eps)

        f1_scores[t_idx]   = f1
        precision_T[t_idx] = p
        recall_T[t_idx]    = r
```

```python
        if verbose:
            print(f"{t:>5.1f}  {f1.item():>8.4f}  {p.item():>8.4f}  {r.item():>8.4f}"
                  f"  {tp.item():>8.0f}  {fp.item():>8.0f}  {fn.item():>8.0f}")

    best_idx       = f1_scores.argmax().item()
    best_f1        = f1_scores[best_idx].item()
    best_threshold = thresholds[best_idx]

    if verbose:
        print("=" * 60)
        print(f"Mejor threshold: t={best_threshold:.1f}")
        print(f"Reconstruction Score (max_t F1): {best_f1:.4f}")
        print("=" * 60)

    return best_f1, best_threshold, f1_scores.cpu().numpy()

print("  Fase 2 rediseñada: barrido T simultáneo + F1 global micro-averaged + max_t")
```

```
  Fase 2 rediseñada: barrido T simultáneo + F1 global micro-averaged + max_t
```

```python
start_time = time.time()

reconstruction_score, best_threshold, f1_per_threshold = reconstruct_boards_gpu(
    sae_features_test,
    bsp_gt_test_pieces,
    hp_mask_TFB,
    active_feature_indices,
    f_max,           # mismo tensor calculado antes de Fase 1
    thresholds=THRESHOLDS,
    verbose=True
)

phase2_time = time.time() - start_time
total_time  = phase1_time + phase2_time

print(f"\nTiempo Fase 2: {phase2_time:.2f} seg")
print(f"Tiempo total:  {total_time:.2f} seg ({total_time/60:.2f} min)")
```

```
============================================================
FASE 2: RECONSTRUCCIÓN - BARRIDO DE THRESHOLDS
============================================================
    t        F1         P         R        TP        FP        FN
------------------------------------------------------------
  0.0    0.0574    0.7531    0.0298     59836     19617   1946164
  0.1    0.0574    0.7531    0.0298     59836     19617   1946164
  0.1    0.0585    0.7561    0.0304     61006     19683   1944994
  0.2    0.0604    0.7630    0.0314     63061     19583   1942939
  0.2    0.0628    0.7680    0.0327     65680     19844   1940320
  0.2    0.0651    0.7707    0.0340     68182     20287   1937818
  0.3    0.0676    0.7727    0.0353     70886     20856   1935114
  0.3    0.0691    0.7720    0.0362     72521     21415   1933479
  0.4    0.0698    0.7665    0.0366     73384     22352   1932616
  0.5    0.0697    0.7608    0.0365     73307     23051   1932693
  0.5    0.0698    0.7558    0.0366     73383     23706   1932617
  0.6    0.0696    0.7490    0.0365     73220     24533   1932780
  0.6    0.0694    0.7363    0.0364     73040     26160   1932960
  0.6    0.0687    0.7264    0.0361     72319     27243   1933681
  0.7    0.0681    0.7107    0.0357     71698     29181   1934302
  0.8    0.0678    0.6956    0.0356     71449     31261   1934551
  0.8    0.0679    0.6746    0.0358     71755     34607   1934245
  0.9    0.0684    0.6534    0.0361     72396     38411   1933604
```

```
  0.9    0.0715    0.6201    0.0379     76087     46610   1929913
  0.9    0.0745    0.5817    0.0398     79856     57426   1926144
  1.0    0.0000    0.0000    0.0000         0         0   2006000
============================================================
Mejor threshold: t=0.9
Reconstruction Score (max_t F1): 0.0745
============================================================

Tiempo Fase 2: 2.73 seg
Tiempo total:  36.98 seg (0.62 min)
```

0.8 8. Análisis de Resultados

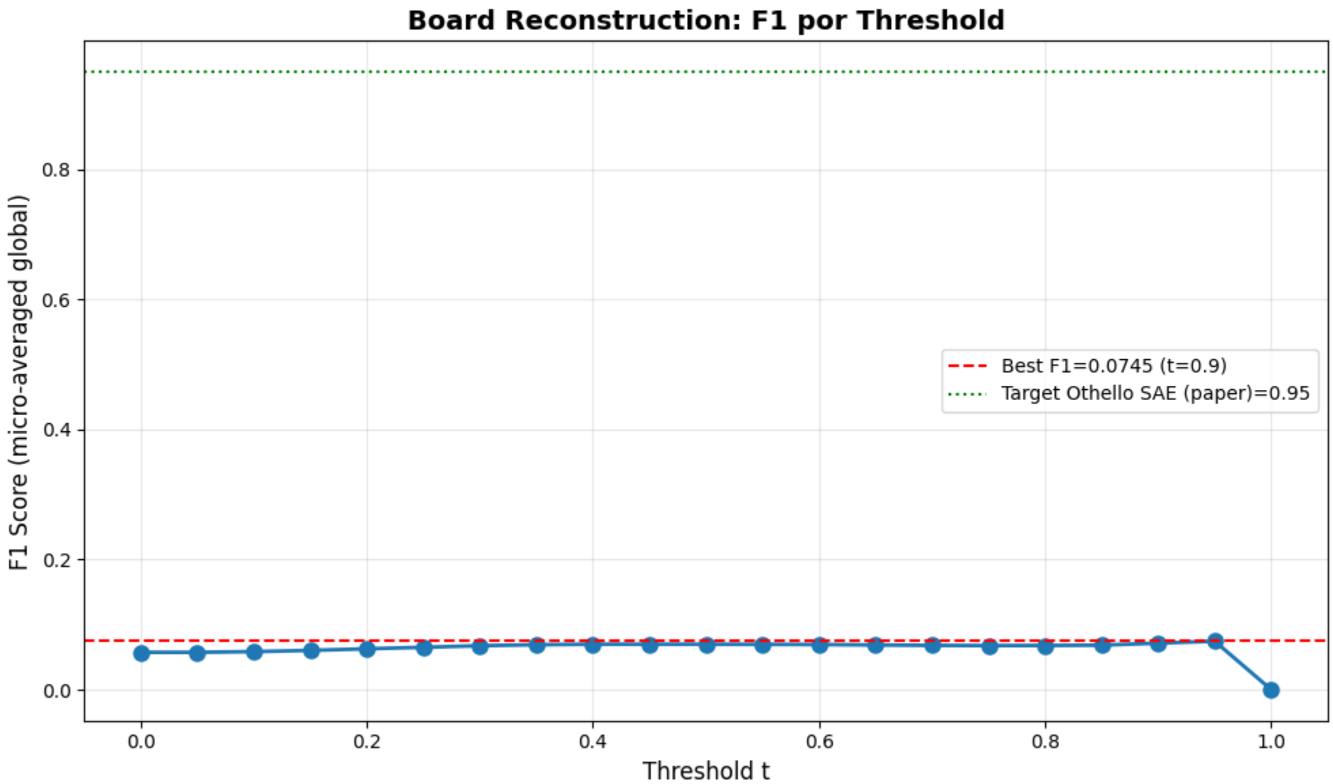
```
print("=" * 60)
print("BARRIDO DE THRESHOLDS - F1 POR THRESHOLD")
print("=" * 60)
for i, (t, f1) in enumerate(zip(THRESHOLDS, f1_per_threshold)):
    marker = " ← MEJOR" if i == f1_per_threshold.argmax() else ""
    print(f"t={t:.1f}: F1={f1:.4f}{marker}")
print("=" * 60)

plt.figure(figsize=(10, 6))
plt.plot(THRESHOLDS, f1_per_threshold, marker='o', linewidth=2, markersize=8)
plt.axhline(y=reconstruction_score, color='r', linestyle='--',
            label=f'Best F1={reconstruction_score:.4f} (t={best_threshold:.1f})')
plt.axhline(y=0.95, color='g', linestyle=':', label='Target Othello SAE (paper)=0.95')
plt.xlabel('Threshold t', fontsize=12)
plt.ylabel('F1 Score (micro-averaged global)', fontsize=12)
plt.title('Board Reconstruction: F1 por Threshold', fontsize=14, fontweight='bold')
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()

print(f"\n  Threshold óptimo: t={best_threshold:.1f}")
print(f"  Reconstruction Score: {reconstruction_score:.4f}")
```

```
============================================================
BARRIDO DE THRESHOLDS - F1 POR THRESHOLD
============================================================
t=0.0: F1=0.0574
t=0.1: F1=0.0574
t=0.1: F1=0.0585
t=0.2: F1=0.0604
t=0.2: F1=0.0628
t=0.2: F1=0.0651
t=0.3: F1=0.0676
t=0.3: F1=0.0691
t=0.4: F1=0.0698
t=0.5: F1=0.0697
t=0.5: F1=0.0698
t=0.6: F1=0.0696
t=0.6: F1=0.0694
t=0.6: F1=0.0687
t=0.7: F1=0.0681
t=0.8: F1=0.0678
t=0.8: F1=0.0679
t=0.9: F1=0.0684
t=0.9: F1=0.0715
t=0.9: F1=0.0745 ← MEJOR
```

```
t=1.0: F1=0.0000
============================================================
```




```
 Threshold óptimo: t=0.9
 Reconstruction Score: 0.0745
```

```python
# Estadísticas sobre features de alta precisión por BSP
hp_per_bsp = hp_mask_TFB.sum(dim=1)  # (T, n_bsps)
best_t_idx = torch.tensor(f1_per_threshold).argmax().item()
n_features_per_bsp = hp_per_bsp[best_t_idx].cpu().numpy()

print("="*60)
print("ESTADÍSTICAS - FEATURES POR BSP")
print("="*60)
print(f"Total BSPs: {len(n_features_per_bsp)}")
print(f"Features promedio por BSP: {np.mean(n_features_per_bsp):.1f}")
print(f"Features mediana por BSP: {np.median(n_features_per_bsp):.1f}")
print(f"BSPs con 0 features: {np.sum(n_features_per_bsp == 0)}")
print(f"BSPs con 1+ features: {np.sum(n_features_per_bsp > 0)}")
print(f"BSPs con 5+ features: {np.sum(n_features_per_bsp >= 5)}")
print(f"BSPs con 10+ features: {np.sum(n_features_per_bsp >= 10)}")
print(f"Max features en una BSP: {np.max(n_features_per_bsp)}")
print("="*60)
```

```
============================================================
ESTADÍSTICAS - FEATURES POR BSP
============================================================
Total BSPs: 128
Features promedio por BSP: 757.6
Features mediana por BSP: 467.0
BSPs con 0 features: 0
BSPs con 1+ features: 128
```

```
BSPs con 5+ features: 128
BSPs con 10+ features: 128
Max features en una BSP: 4048
============================================================
```

0.9 9. Guardar Resultados

```
results = {
    'reconstruction_score': reconstruction_score,
    'best_threshold': best_threshold,
    'f1_per_threshold': f1_per_threshold,
    'thresholds': np.array(THRESHOLDS),
    'phase1_time_seconds': phase1_time,
    'phase2_time_seconds': phase2_time,
    'total_time_seconds': total_time,
    'bsp_names': bsp_pieces_names,
}

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)
output_path = metrics_dir / f"reconstruction_results_{filter_suffix}.npz"
np.savez(output_path, **results)

print(f"  Resultados guardados en: {output_path}")
print(f"\nContenido:")
print(f"  reconstruction_score: {reconstruction_score:.4f}")
print(f"  best_threshold:       {best_threshold:.1f}")
print(f"  f1_per_threshold:     {f1_per_threshold}")

# Comparación con el paper
print()
print("=" * 60)
print("COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)")
print("=" * 60)
print(f"{'Modelo':<25} {'Reconstruction':>14}")
print("-" * 40)
print(f"{'SAE trained (paper)':<25} {'0.95':>14}  ← Objetivo Othello")
print(f"{'Nuestro SAE':<25} {reconstruction_score:>14.4f}  ← Resultado")
print("=" * 60)

if reconstruction_score >= 0.90:
    print("\n  Excelente: Muy cercano al objetivo del paper")
elif reconstruction_score >= 0.50:
    print("\n~ Moderado: Por encima de random, revisar SAE")
else:
    print("\n  Bajo: Revisar entrenamiento del SAE o calidad de las activaciones")
```

```
 Resultados guardados en: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_batch-
topk_topk\100000games\ex2\metrics\reconstruction_results_absolute.npz

Contenido:
  reconstruction_score: 0.0745
  best_threshold:       0.9
  f1_per_threshold:     [0.05738418 0.05738418 0.05847158 0.06038463 0.06280588 0.06510672 0.06758315 0.06906972 0.0698
```

```
 0.06973789 0.06978592 0.06960893 0.06939009 0.0686933  0.06806087 0.0677656  0.06793816 0.06840114
 0.07148692 0.0745175  0.        ]

============================================================
COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)
============================================================
Modelo                    Reconstruction
----------------------------------------
SAE trained (paper)                 0.95  ← Objetivo Othello
Nuestro SAE                       0.0745  ← Resultado
============================================================

  Bajo: Revisar entrenamiento del SAE o calidad de las activaciones
```

0.10 10. Generar PDF con Quarto

```python
output_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    f'reconstruction_{filter_suffix}',
    extras_id=extras_id
)

notebook_name = 'reconstruction_sae_batchtopk.ipynb'
output_path = output_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{output_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)
```

```
 Generando PDF con Quarto...
 Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_batch-
topk_topk\100000games\ex2\metrics\sae_topk_batch-topk_l6_100000g_ex2_reconstruction_absolute.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_batch-
topk_topk\100000games\ex2\metrics\sae_topk_batch-topk_l6_100000g_ex2_reconstruction_absolute.pdf
```